

Project Charta

Problem Definition

Have you ever been to a restaurant or in a car where a song is playing that you just can't quite identify? Whether it's a song you recognize but don't know the name for, or it's a new song you found that you really like, the solution is simple: Ampli-FIRE. With Ampli-FIRE's song identifying technology all the user has to do is upload a snippet of the song and the software will identify it using its own extensive database. Even if the song is not currently in the database, Ampli-FIRE will identify the song's genre and output the four songs it thinks you will like best from a list of 545,008 songs.

Stakeholders: Anna Baranek(developer), My-Hanh Dang(developer), Kalia Dorgelo(developer), Liz Stahlbaum(developer)

Situation Assessment

Resources: python(language), streamlit(UI)

Project Goals and Success Criteria

If a user chooses to upload an audio file the program will output the song name and author(if it is in the database) and/or the predicted genre of the given song. Using the predicted genre, the program will then output a list of 4 songs that are similar to the given one.

If a user chooses to input a song name, artist, and genre then the program will output 4 songs that are similar to the given one.

Parameters for Success

- Users are given the correct song they are looking for
- Users are given songs that are similar to the given song
- Seamless user interface that is easy for all users to use

---Ampli-FIRE is not responsible for predicting song preferences using any information beyond the singular song provided as we do not want to store any user data at this time---

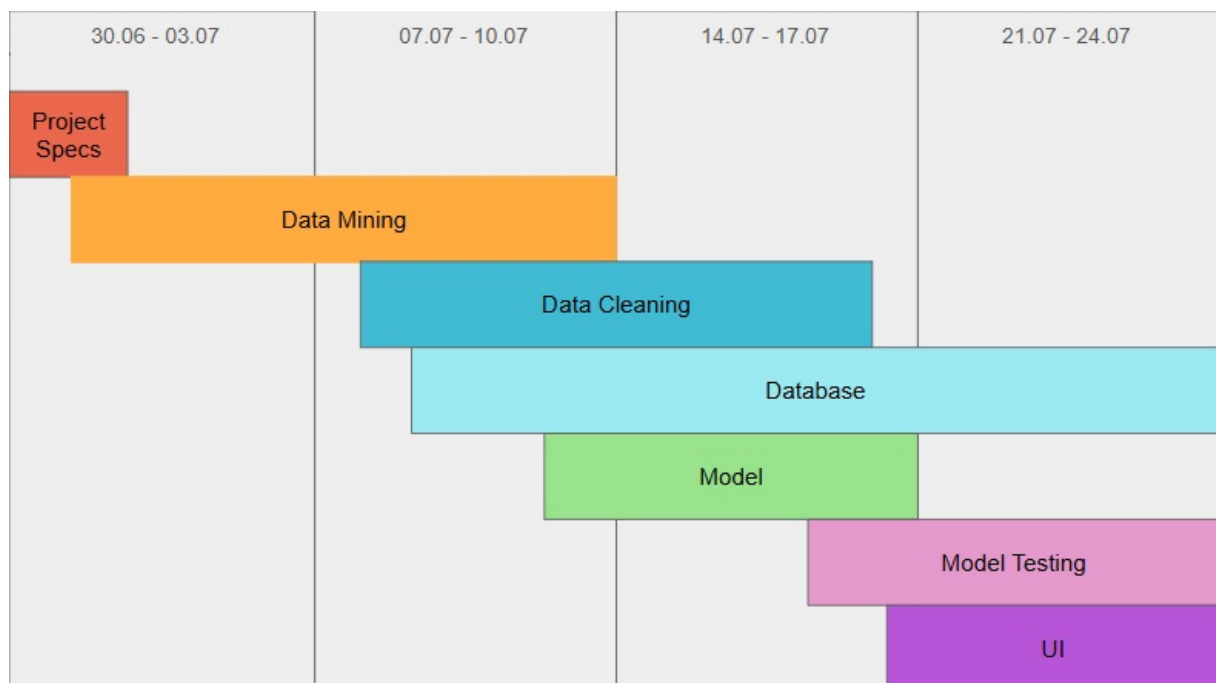
Data Mining Goals

The team hopes to continuously add to the database, providing as many songs of as many genres as possible. This will be done in hopes of allowing for more song matches and more accurate recommendations.

---Problem(s): An extensive amount of songs fall under copyright laws making it impossible to store the audio files/fingerprints needed for the song matching portion of the application---

Project Plan

Divide the project into individual phases, describe them briefly and draw up a preliminary timetable, e.g. as a Gantt chart:



Roles and Contact Details

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